THUE'S EQUATION

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This worksheet works through [Sil09, Exercises 9.5 and 9.6(a)–(b)]. In 1909 Thue proved that an equation $F(x, y) = b$, where F is a binary form of degree at least 3 with integer coefficients and nonzero discriminant and b is a nonzero integer, has only finitely many integer solutions. His idea was to turn the equation into a problem of Diophantine approximation. A solution with y large makes x/y very close to a root of $F(T, 1)$, and algebraic numbers cannot be approximated too well by rational numbers. Problem 1.6 makes the first step quantitative: a polynomial is small only near its roots. Problem 2.5 proves Thue's theorem with Roth's theorem, and extends it to the S-integers of a number field for forms with an irreducible factor of degree at least 3. One part of each exercise is corrected or restricted, as explained at the start of each section.

Standing notation.

- A number field K, its set of places $\mathcal{P}_K$, and the subset $\mathcal{P}_K^\infty$ of archimedean places.
- For $v \in \mathcal{P}_K$, the completion K_v, the absolute value $|\ |_v$ on K_v extending the usual absolute value on $\mathbb{Q}_p$ or $\mathbb{R}$, and the local degree $n_v \coloneqq [K_v : \mathbb{Q}_v]$, where $\mathbb{Q}_v$ is the completion of $\mathbb{Q}$ at the restriction of v.
- An algebraic closure $\overline{K}$ of K. For $v \in \mathcal{P}_K$, an algebraic closure $\overline{K_v}$ of K_v, to which $|\ |_v$ extends uniquely, as K_v is complete, and a K-embedding $\iota_v \colon \overline{K} \to \overline{K_v}$. Then $|z|_v \coloneqq |\iota_v(z)|_v$ extends $|\ |_v$ to $\overline{K}$. The ι_v are arbitrary, and every statement below holds for every choice of them. For $K = \mathbb{Q}$, the field $\overline{\mathbb{Q}}$ is the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$, ι_∞ is the inclusion $\overline{\mathbb{Q}} \subseteq \mathbb{C}$, and $|\ |$ is the usual absolute value.
- The height $H_K(P) \coloneqq \prod_{v \in \mathcal{P}_K} \max_i |x_i|_v^{n_v}$ of a point $P = [x_0 : \cdots : x_N]$ of $\mathbb{P}^N(K)$, which does not depend on the coordinates and satisfies $H_K(P) \geqslant 1$ [Sil09, §VIII.5]. For $x \in K$, $H_K(x) \coloneqq H_K([x : 1])$. For coprime integers p and q with $q \neq 0$, $H_{\mathbb{Q}}(p/q) = \max(|p|, |q|)$.
- The product formula $\prod_{v \in \mathcal{P}_K} |a|_v^{n_v} = 1$ for $a \in K^\times$ [Sil09, Product Formula VIII.5.3].

How to use this. Allow about 30 minutes for Problem 1.6 and 35 for Problem 2.5. Problem 2.5 uses only Problem 1.6(1) and (3), so Problem 1.6(4) and (5) can wait for a second pass. The `Magma` block runs in seconds in the free online calculator. Everything else is by hand.

1. Polynomials are small only near their roots

Problem 1.6 is [Sil09, Exercise 9.5]. If t is far from every root of a polynomial f, then $|f(t)|$ is not small. Quantitatively, $|f(t)|$ is at least a constant times the distance from t to the nearest root, and the constant depends on how close the roots are to each other, which the discriminant controls. Silverman's last part asks for a constant that depends only on the degree of f and on the projective height of its vector of coefficients. No such constant exists: dividing f by an integer N leaves this height unchanged, but at an archimedean place it divides $|f(t)|_v$ by N. So Problem 1.6(5) asks for a constant that depends on the affine height of the coefficients instead.

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Definition 1.1 (Discriminant)**.** Let L be a field and $f = a_0T^n + a_1T^{n-1} + \cdots + a_n \in L[T]$ with $n \geqslant 1$ and $a_0 \neq 0$. Over an algebraic closure of L, $f = a_0 \prod_{i=1}^n (T - \xi_i)$. The discriminant of f is

$$D(f) \coloneqq a_0^{2n-2} \prod_{i<j} (\xi_i - \xi_j)^2.$$

It is a polynomial with integer coefficients in $a_0, \ldots, a_n$, by the theorem on symmetric polynomials, since each ξ_i occurs in $\prod_{i<j}(\xi_i - \xi_j)^2$ to degree at most $2n-2$. So $D(f) \in L$, and $D(f) \in \mathbb{Z}$ when $f \in \mathbb{Z}[T]$. It is nonzero if and only if the roots of f are distinct.

Definition 1.2 (Mahler measure [HS00, §B.7])**.** The Mahler measure of $f = a_0 \prod_{i=1}^n (T - \xi_i) \in \mathbb{C}[T]$, with $a_0 \neq 0$, is

$$M(f) \coloneqq |a_0| \prod_{i=1}^n \max(1, |\xi_i|).$$

Lemma 1.3 (Mahler measure and coefficients [HS00, Lemma B.7.3.1 and the formula after it])**.** *Let $f = a_0T^n + \cdots + a_n \in \mathbb{C}[T]$ with $a_0 \neq 0$. Then $M(f) \leqslant (n+1)^{1/2} \max_k |a_k|$.*

Proposition 1.4 (Gauss's lemma [HS00, proof of Proposition B.7.3])**.** *Let v be a non-archimedean place of K. For a polynomial g with coefficients in $\overline{K}$, let $|g|_v$ be the maximum of the absolute values of its coefficients. Then $|gh|_v = |g|_v |h|_v$. In particular, if $f = a_0 \prod_{i=1}^n (T - \xi_i)$, then*

$$|f|_v = |a_0|_v \prod_{i=1}^n \max(1, |\xi_i|_v).$$

Proposition 1.5 (Hadamard's inequality)**.** *For $z = (z_1, \ldots, z_m) \in \mathbb{C}^m$, let $\|z\| \coloneqq \left(\sum_k |z_k|^2\right)^{1/2}$. Let V be an $m \times m$ complex matrix with rows $r_1, \ldots, r_m$. Then $|\det V| \leqslant \prod_{i=1}^m \|r_i\|$.*

Problem 1.6 (Polynomials are small only near their roots)**.**
Let $f = a_0T^n + a_1T^{n-1} + \cdots + a_n \in K[T]$ be a polynomial of degree $n \geqslant 2$, with $a_0 a_n \neq 0$ and with distinct roots $\xi_1, \ldots, \xi_n \in \overline{K}$. So $f = a_0 \prod_{i=1}^n (T - \xi_i)$, and $D(f) \neq 0$. (Silverman allows $n = 1$, where $f(t) = a_0(t - \xi_1)$ and every statement below is immediate.)

(1) Let $v \in \mathcal{P}_K$, and let $\varepsilon_v \coloneqq 1/2$ if v is archimedean and $\varepsilon_v \coloneqq 1$ if v is non-archimedean. Let $t \in \overline{K}$, and let ξ_i be a root of f nearest to t, that is, $|t - \xi_i|_v = \min_j |t - \xi_j|_v$. Show that

$$|f(t)|_v \geqslant \varepsilon_v^{n-1} |f'(\xi_i)|_v |t - \xi_i|_v.$$

a

(2) In this item and the next, $K = \mathbb{Q}$, $f \in \mathbb{Z}[T]$, and $|\ |$ is the usual absolute value on $\mathbb{C}$. Show that every root ξ_i of f satisfies

$$|f'(\xi_i)| \geqslant (n-1)^{-(n-1)/2} M(f)^{-(n-2)}.$$

b

(3) Let $A \coloneqq \max(|a_0|, \ldots, |a_n|)$. Show that every rational number t satisfies

$$|f(t)| \geqslant \frac{\min\big(|t - \xi_1|, \ldots, |t - \xi_n|\big)}{2^{n-1}(n+1)^{n-3/2} A^{n-2}}.$$

(4) Let $S \subseteq \mathcal{P}_K$ be a finite set of places. Show that there is a constant $C_f > 0$, depending only on f, such that every $t \in K$ satisfies

$$\prod_{v \in S} \min\big(1, |f(t)|_v^{n_v}\big) \geqslant C_f \prod_{v \in S} \min_{1 \leqslant i \leqslant n} \min\big(1, |t - \xi_i|_v^{n_v}\big).$$

(5) Let $H \coloneqq H_K([1 : a_0 : \cdots : a_n])$. Find a constant C_f for (4) that depends only on n, $[K : \mathbb{Q}]$ and H.[c]

[a] *Hint:* for $j \neq i$, compare $|\xi_i - \xi_j|_v$ with $|t - \xi_j|_v$.
[b] *Hint:* number the roots so that $i = 1$, write $D(f)$ as $f'(\xi_1)^2$ times the square of $a_0^{n-2} \prod_{2 \leqslant i < j \leqslant n} (\xi_i - \xi_j)$, and apply Hadamard's inequality to the Vandermonde matrix of $\xi_2, \dots, \xi_n$.
[c] *Hint:* for $v \in S$, bound $|f'(\xi_i)|_v$ from below as in (2), with Gauss's lemma in place of the Mahler measure at the non-archimedean places. Then bound $\prod_{v \in S} \min(1, |D(f)|_v^{n_v})$ from below by $H_K(D(f))^{-1}$.

2. Thue's equation

Problem 2.5 is [Sil09, Exercise 9.6(a)–(b)], with Roth's theorem in the form of [HS00, Theorem D.2.1]. If y is large, then at each place of S the ratio x/y of a solution is close to a root of $F(T, 1)$, and this root may change from place to place. Theorem 2.3 has a single algebraic number ξ, but it holds for every choice of the embeddings ι_v, so ξ can stand for a different conjugate at each place. So Problem 2.5(3) assumes that F has an irreducible factor of degree at least 3, whose roots are conjugate. Silverman states that part for every F of degree at least 3 with nonzero discriminant.

Definition 2.1 (Binary forms)**.** A binary form of degree d over K is a nonzero homogeneous polynomial $F \in K[X, Y]$ of degree d. Over $\overline{K}$ it is a product of linear forms,

$$F = \prod_{i=1}^{d} L_i, \qquad L_i = \beta_i X - \alpha_i Y.$$

Let $\Delta_{ij} \coloneqq \alpha_i \beta_j - \alpha_j \beta_i$. The discriminant of F is $\prod_{i<j} \Delta_{ij}^2$. It does not depend on the factorization, and if $F(1, 0) \neq 0$, it is the discriminant $D(f)$ of $f(T) \coloneqq F(T, 1)$. It is nonzero if and only if no two of the L_i are proportional.

Definition 2.2 (S-integers [Sil09, proof of Proposition VIII.1.6])**.** Let $S \subseteq \mathcal{P}_K$ be a finite set containing $\mathcal{P}_K^\infty$. The ring of S-integers of K is

$$\mathcal{O}_S \coloneqq \{a \in K : |a|_v \leqslant 1 \text{ for all } v \notin S\}.$$

Theorem 2.3 (Roth [HS00, Theorem D.2.1])**.** *Let $S \subseteq \mathcal{P}_K$ be a finite set, $\xi \in \overline{K}$ and $\varepsilon > 0$. Then only finitely many $t \in K$ satisfy*

$$\prod_{v \in S} \min\bigl(1, |t - \xi|_v^{n_v}\bigr) \leqslant H_K(t)^{-(2+\varepsilon)}.$$

Theorem 2.4 (Northcott [Sil09, Theorem VIII.5.11])**.** *For every $B > 0$, only finitely many $t \in K$ satisfy $H_K(t) \leqslant B$.*

Problem 2.5 (Thue's equation)**.**
Let $F \in K[X, Y]$ be a binary form of degree $d \geqslant 3$ with nonzero discriminant, and let $b \in K^\times$.
(1) Let $K = \mathbb{Q}$, and suppose that $F \in \mathbb{Z}[X, Y]$ and $b \in \mathbb{Z}$. Show that $F(x, y) = b$ has only finitely many solutions $(x, y) \in \mathbb{Z}^2$.[a]
(2) `Magma` solves Thue equations. The code below lists all the integer solutions of $x^3 - 2y^3 = b$ for $1 \leqslant b \leqslant 6$. Explain why the argument of (1) could not produce such a list.

```
R<x> := PolynomialRing(Integers());
t := Thue(x^3 - 2);                    // the form X^3 - 2*Y^3
[ <b, Solutions(t, b)> : b in [1..6] ];
// [ <1, [[-1, -1], [1, 0]]>, <2, [[0, -1]]>, <3, [[-5, -4], [1, -1]]>,
```

```
//   <4, []>, <5, []>, <6, [[2, 1]]> ]
```

(3) Let $S \subseteq \mathcal{P}_K$ be a finite set containing $\mathcal{P}_K^\infty$, and suppose that F has an irreducible factor $G \in K[X,Y]$ of degree $g \geqslant 3$. Show that $F(x,y) = b$ has only finitely many solutions $(x,y) \in \mathcal{O}_S \times \mathcal{O}_S$.[b]

[a] *Hint:* first reach $F(1,0)F(0,1) \neq 0$ with changes of variables $(x,y) \mapsto (x, y+mx)$ and $(x,y) \mapsto (x+my, y)$, $m \in \mathbb{Z}$. Then let $f(T) = F(T,1)$, write $b = y^d f(x/y)$ for $y \neq 0$, and use Problem 1.6(3) and Roth's theorem.

[b] *Hint:* reduce to $F(1,0) \neq 0$ as in (1), and let $\xi_1, \dots, \xi_g$ be the roots of $G(T,1)$. For a solution with $y \neq 0$, use that F/G evaluated at (x,y) is a nonzero S-integer, after enlarging S, and the product formula to bound $\prod_{v \in S} \min_i \min(1, |x/y - \xi_i|_v^{n_v})$ by a constant times $H_K(x/y)^{-g}$. Then change the embeddings ι_v so that one root is the nearest at every $v \in S$.

References

[HS00] Marc Hindry and Joseph H. Silverman, *Diophantine geometry*, Graduate Texts in Mathematics, vol. 201, Springer-Verlag, New York, 2000. An introduction. MR1745599 ↑

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑